Organic Cume 9/14/2012

1. Draw and give relative energies of methane's occupied molecular orbitals. Show how they are constructed from atomic orbitals.

2. Why is CH_3^+ more accurately called a carbenium ion?

3. Discuss the stability of the allyl carbocation. Relate its stability to that of primary, secondary, and tertiary carbocations.

4. Discuss why a tertiary carbocation is more stable than a primary carbocation. Rationalize this using both Valence Bond Theory and Molecular orbital theory.

5. Why is *trans*-2-butene more stable than 1-butene?

6. Discuss the bonding and structure of the intermediate methylene "CH_2".